AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Project Report

Project Title:	**Expense Management System**		
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Course Title:	Programming in Python		
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Semester:	Spring 2025-26	Course Teacher:	**MD. Tanzeem Rahat**

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Expense Management System – Project Report

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1. Introduction

An Expense Management System has been developed as a desktop application using the Python programming language to assist users in effectively managing their daily expenses. This system is intended to assist users in maintaining better control over their spending habits. A graphical user interface has been incorporated to make the system more user-friendly. Visualization techniques have also been incorporated to better comprehend the spending patterns. The application is an example of the application of various programming concepts.

2. Objectives

The following objectives have been considered during the development of this project:

- To design a simple and intuitive expense tracking system
- To implement core operations such as adding, editing, deleting, and searching records
- To provide summarized insights of expenses
- To incorporate graphical visualization for better understanding
- To apply object-oriented programming concepts in Python

3. Features

The system includes the following functionalities:

- Addition of new expense records
- Modification of existing entries
- Deletion of selected expenses
- Searching expenses based on category
- Sorting expenses by amount (ascending and descending)
- Display of summary information including total expense and top category
- Visualization through pie charts and bar charts
- Interactive and modern graphical user interface

4. System Architecture

The application has been organized into multiple components to ensure clarity and maintainability.

User Interface Layer

The interface has been developed using CustomTkinter and Tkinter components. It manages user inputs, displays data in a structured format, and handles interaction with the system.

Logic Layer

A separate manager module has been used to handle all operations related to expense data, including storage, updates, searching, sorting, and generating summaries.

Data Structure

Each expense entry consists of:

- Category
- Amount
- Date
- Additional note

This separation of concerns ensures that the interface remains independent from the core logic.

5. System Screenshots

5.1 Main Interface

The primary interface of the application is shown below. It includes input fields, control buttons, and a table for displaying stored expense records.

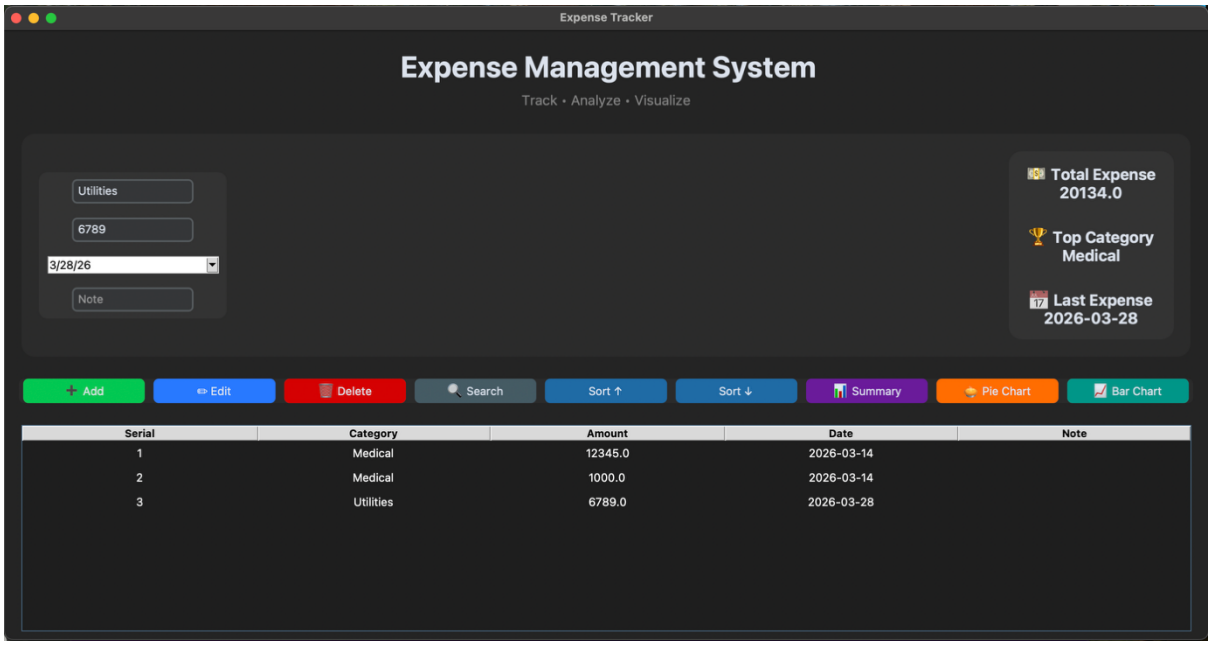

Figure 1: Main Interface

5.2 Add Expense Functionality

The process of adding a new expense entry is illustrated below. The required details are entered and subsequently reflected in the table.

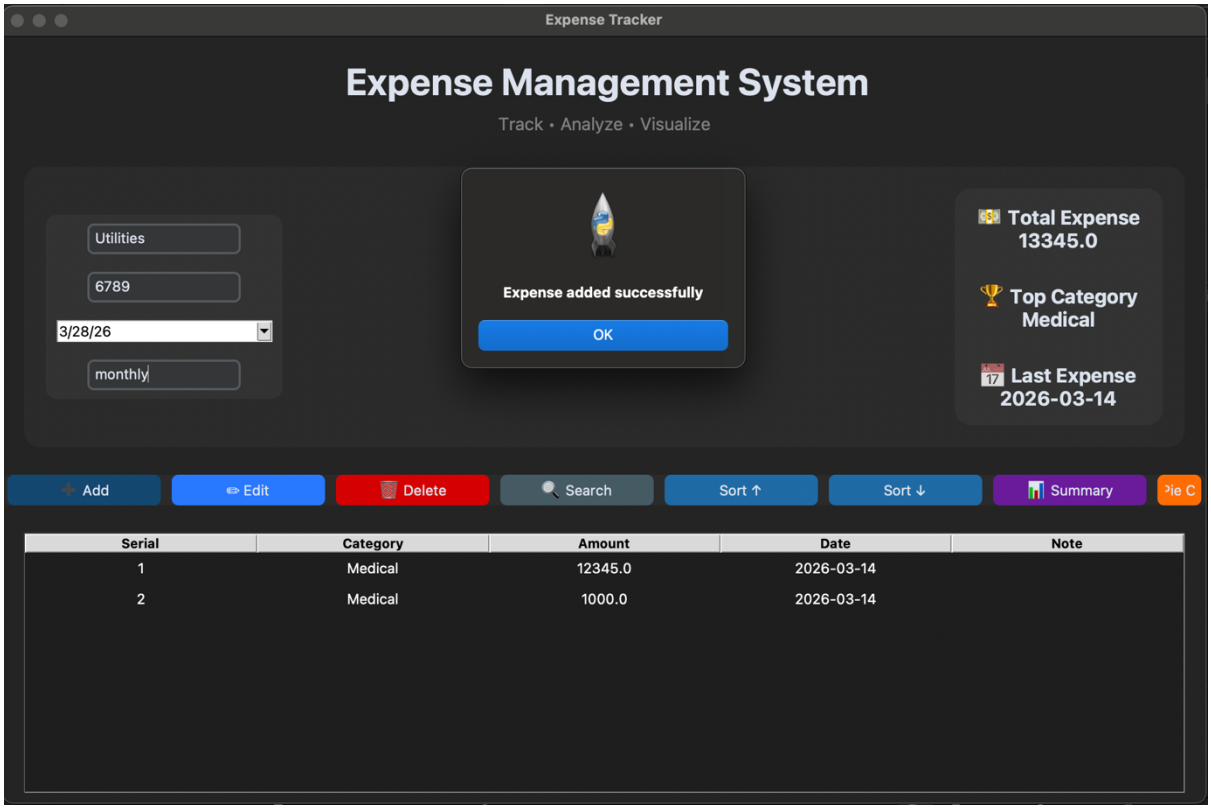

Figure 2: Add Expense Functionality

5.3 Edit and Delete Operations

The editing and deletion features are demonstrated below. A selected entry can be modified or removed as required.

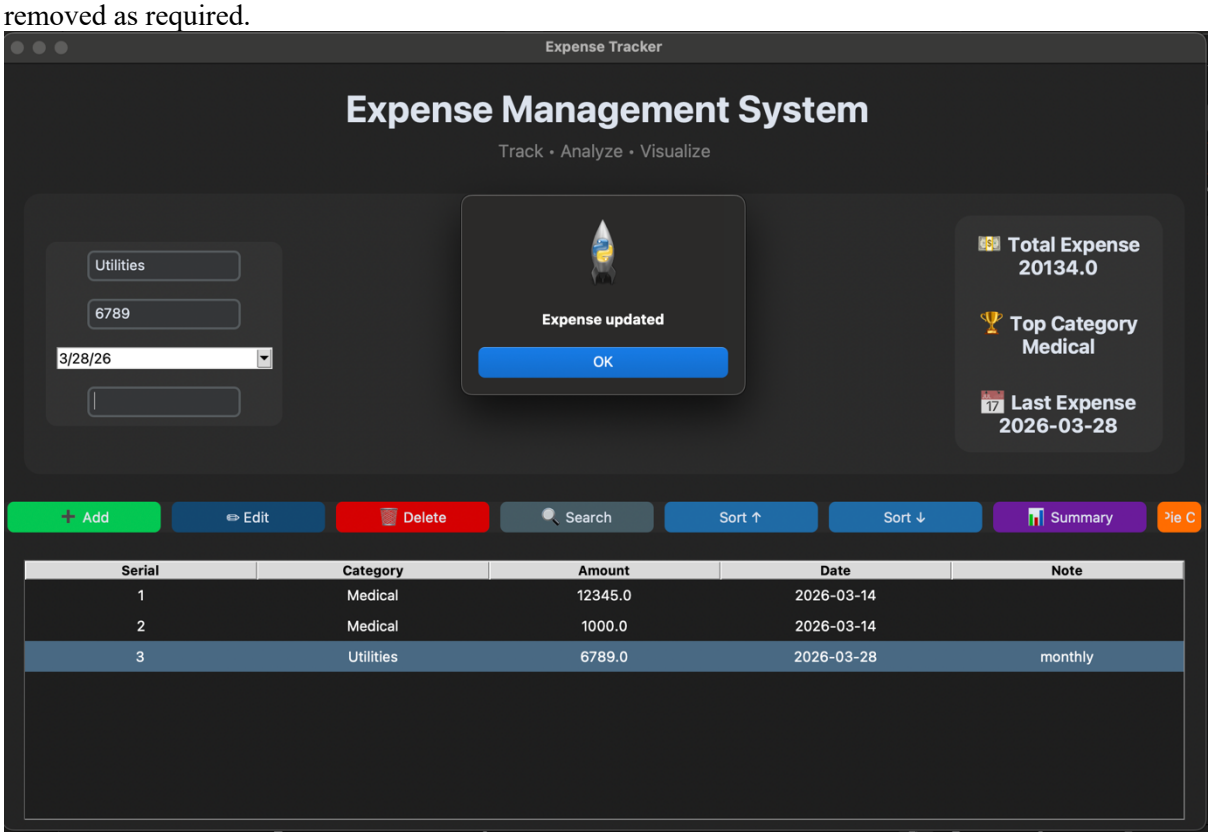

Figure 3.1: Edit Operations

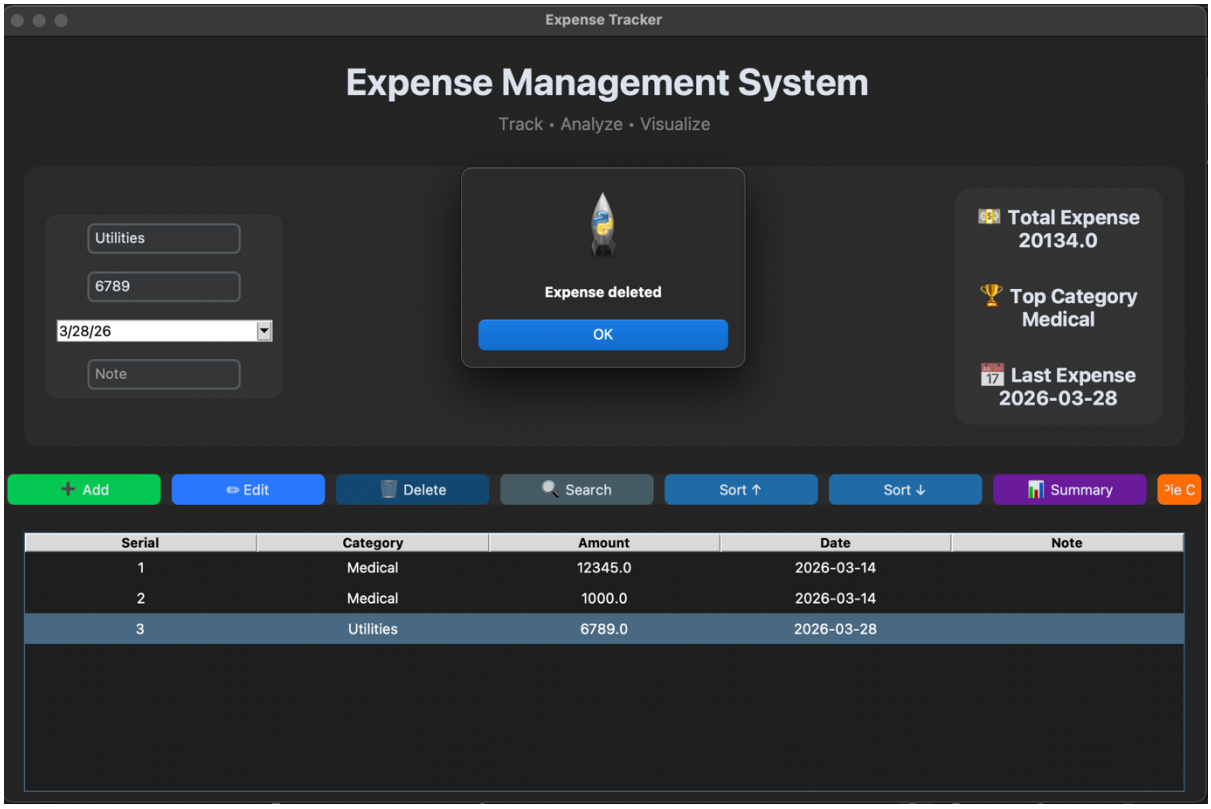

Figure 3.2: Delete Operations

5.4 Summary Dashboard

The summary section is presented below, where total expenses, the highest spending category, and the most recent entry are displayed.

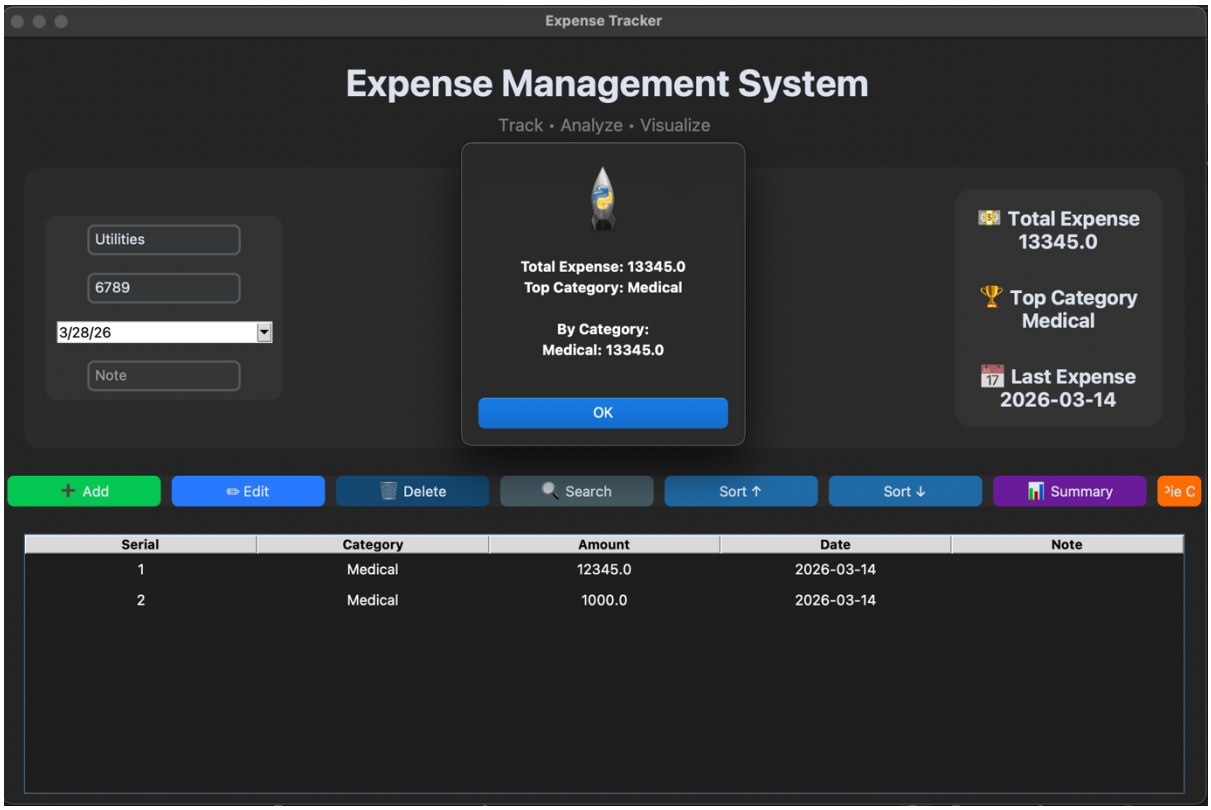

Figure 4: Summary Dashboard

5.5 Pie Chart Visualization

A graphical representation of expenses by category is shown below in the form of a pie chart.

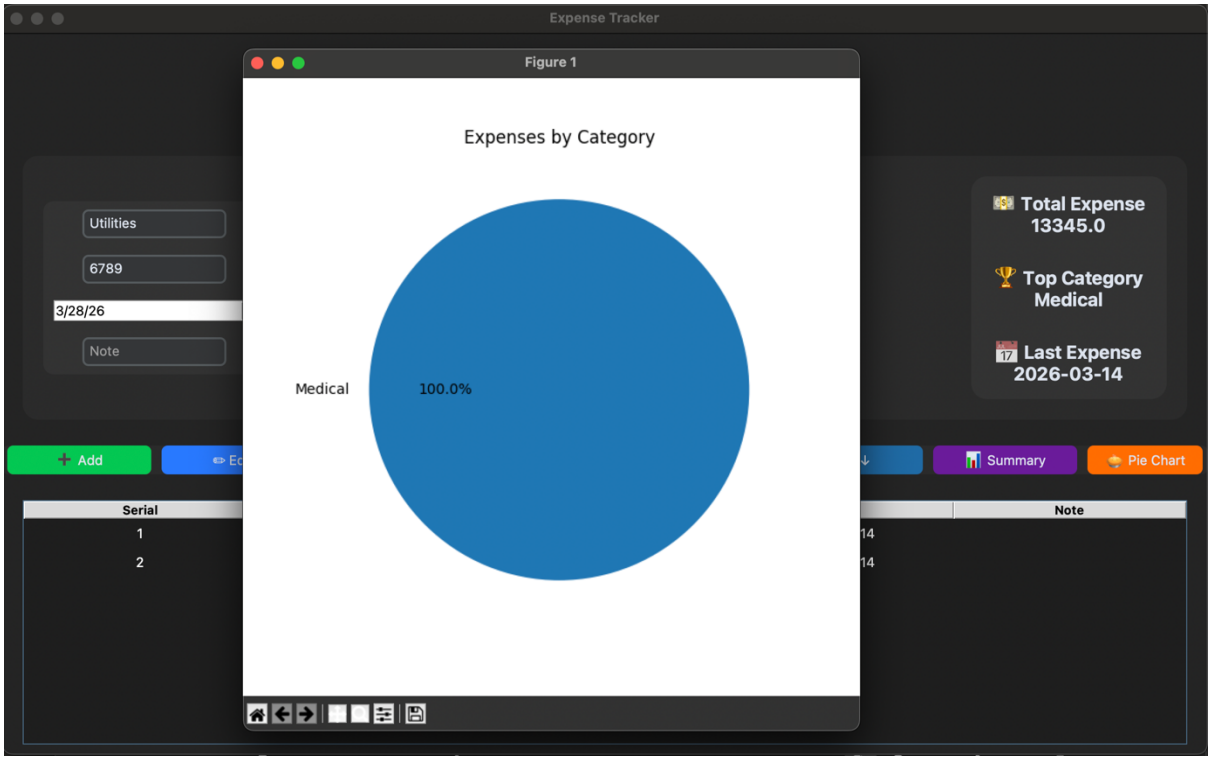

Figure 5: Pie Chart Visualization

5.6 Bar Chart Visualization

The monthly trend of expenses is illustrated below using a bar chart, allowing for easier comparison across different time periods.

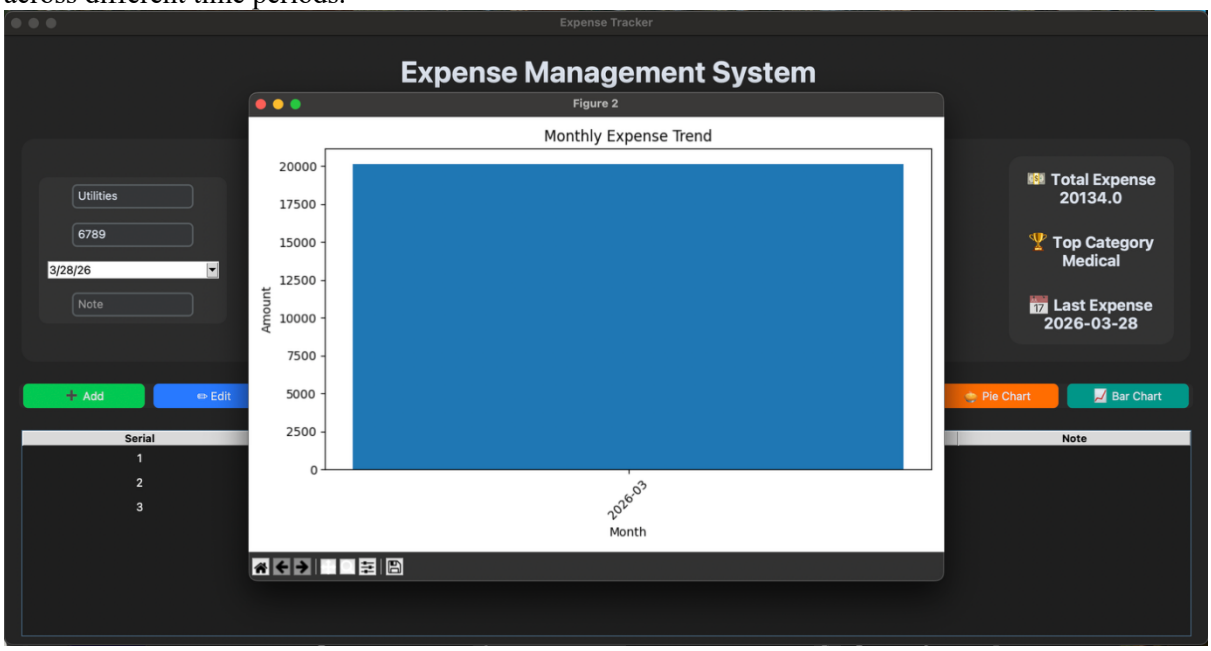

Figure 6: Bar Chart Visualization

6. Installation and Setup

The following steps may be followed to run the project:

Step 1: Clone the Repository

```
git clone https://github.com/Agdam-Amin-Adib/python-midterm-project
cd python-midterm-project/Agdam
```

Step 2: Create a Virtual Environment

```
python -m venv .venv
```

Step 3: Activate the Virtual Environment

- Windows:

```
.venv\Scripts\activate
```

- macOS/Linux:

```
source .venv/bin/activate
```

Step 4: Install Required Packages

```
pip install -r requirements.txt
```

Step 5: Run the Application

```
python main.py
```

(Alternatively, python app.py may be used depending on the entry file.)

Step 6: Freeze Dependencies (Optional)

```
pip freeze > requirements.txt
```

7. System Workflow

The system operates through a sequence of interactions between the user interface and the underlying logic:

1. Expense details are entered through the interface
2. The data is processed by the manager module
3. Records are stored and updated accordingly
4. The interface refreshes to display the latest data
5. Summary and visualization features provide insights into the recorded expenses

8. Key Functional Modules

Add Expense

New expense data is collected and stored after validation.

Edit Expense

Existing records can be updated based on user input.

Delete Expense

Selected entries are removed from the system.

Search Functionality

Expenses can be filtered using category-based queries.

Sorting

Data can be arranged based on expense amount.

Summary

A concise overview of total and categorized expenses is generated.

Visualization

Charts are produced to represent data in graphical form.

9. Limitations

Some limitations have been identified in the current version:

- Data is stored temporarily (no database integration)
- Input validation can be further improved
- Sorting modifies the original data order
- No authentication mechanism is included

10. Future Improvements

Several enhancements may be considered for future development:

- Integration with a database system (e.g., SQLite or MySQL)
- Implementation of user authentication
- Export functionality (CSV/Excel)
- Improved input validation and error handling
- Enhanced user interface responsiveness

11. Conclusion

The project of Expense Management System has been successfully implemented with the desired features and functionalities. This project also shows the efficient use of Python in GUI development and visualization of data.

The project offers a practical solution to managing expenses and also shows a good understanding of programming concepts. The project has the potential to be developed into a more comprehensive financial management system with further enhancements.

12. References

- Python Official Documentation
- CustomTkinter Documentation
- Matplotlib Documentation
- Tkinter and ttk Documentation

The End